

MARCO ANTONIO FAGANELLO

Consultant and Data Scientist

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Introduction

As a Doctor in Political Science from UNICAMP, with a research period at MIT, I bring a robust academic background with a strong emphasis on quantitative analysis and data science. With 7 years of experience in data analysis consulting, I am the founder of MAF dataScience and VotoMaps. My expertise lies in transforming complex data into actionable insights for strategic decision-making.

Skills

- **Programming Languages:** Python (Pandas, NumPy, Scikit-learn, TensorFlow/Keras, PyTorch), R (tidyverse, caret), SQL.
- **Data Analysis:** Statistical Analysis (Quantitative and Qualitative), Geospatial Analysis (GIS, georeferenced data), Web Scraping.
- **Visualization & Tools:** Power BI, Google Looker Studio (formerly Google Data Studio), Gephi, Microsoft Excel.
- **Artificial Intelligence:** Machine Learning (supervised and unsupervised models), Deep Learning (Deep Neural Networks), Natural Language Processing (NLP) with a focus on LLMs and BERT.
- **Methodologies & Capabilities:** Hypothesis generation, data modeling, creation of reports (customized and automated) and interactive dashboards.
- **Strategic Consulting:** Transforming data into actionable insights for decision-making, consulting on the evaluation and analysis of social and public projects.
- **Languages:** Fluent English and Native Portuguese.

Areas of Expertise

My experience spans a broad spectrum of domains, with a focus on applying data science to generate strategic insights:

Electoral & Political Analysis:

- Electoral consulting and mapping for campaigns (Federal and State Deputies, Councilors, Mayors), including geospatial voting analysis (São Paulo, Belém).
- Studies on the impact of Electoral Management Bodies (EMBs) on democratic quality (University of Notre Dame), analysis of specific candidacies (police officers), and dynamics of national and global elections (CEPESP Data, ESEB, CSES).
- Monitoring and analysis of political behavior on social media.

Applied Social Sciences & Public Policy:

- Research and data analysis on critical social issues such as **Child Labor**, **School Dropout** (Aprendiz, Fundação Vale), **Forced Labor** (FGV), and **governmental actions in public health** (COVID-19 Observatory, Miami University).
- Execution and analysis of **surveys** (street vendors Dieese, small merchants in Mozambique IPSOS, Banco do Brasil employees), including questionnaire programming (xlsform) and real-time field monitoring.
- Development and management of complex databases for academic and applied research.

Data Consulting & Strategy:

- Creation and management of complex and **georeferenced databases** (Churches, Caixa Bank properties, electoral data).
- **Web Scraping** for massive data collection (Caixa Bank properties, FAPESP, Twitter).
- Development of **automation solutions** for communication (WhatsApp, email) and data collection (automatic dialing applications).
- **Data Visualization & Reporting:** Production of interactive dashboards, online electoral maps (VotoMaps), and customized reports for various purposes.

Applied Artificial Intelligence & Machine Learning:

- Development of **Deep Learning models** (Deep Neural Networks) for data classification, including **fine-tuning of pre-trained models (BERT)** for accuracy optimization.
- Application of advanced techniques for categorization and extraction of insights from textual and structured data.

Experience

July 2025 | Street Vendor Survey - Dieese

Conducted an in-person survey with 3,000 street vendors in the city of São Paulo. I was responsible for programming the questionnaire in XLSForm, adapting questionnaires to devices, real-time field monitoring with dashboards, and updating and maintaining the database.

April 2025 | Caixa Bank Property Web Scraping - Real Estate Company

Developed a Web Scraping algorithm for cataloging property data from Caixa Econômica Federal. The program was developed in Python with browser automation.

April 2025 | Elections Around the World - University of Notre Dame

Created a database and conducted analysis on the impact of EMBs (Electoral Management Bodies) on the quality of democracies worldwide. The study was performed with analysis using R.

February 2025 | Deep Neural Network Classification - CEPESPDData

Developed a Deep Neural Network model for categorizing candidate assets, utilizing a pre-trained BERT model with supervised learning. Optimization of classification was achieved through fine-tuning and hyperparameter adjustment, enhancing accuracy in automatic categorization.

October 2024 | Church Database - University of Texas at Austin

Constructed a geolocated database of Catholic churches across the national territory, made available in RData and CSV formats for geospatial analysis and visualization.

January 2024 | Postdoctoral Research - FGV/EAESP/CEPESP

Conducted research on Brazilian elections and politics at the Center for Public Sector Politics and Economics of Fundação Getúlio Vargas. Involved in the administration and management of the CEPESP Data and RAIS electoral databases.

October 2023 | Small Merchant Research - IPSOS - Mozambique

Reviewed and analyzed survey data from small merchants in Mozambique. This involved reviewing data in SPSS codes and translating it to R.

July 2023 | ESEB 2022 - CESOP

Analyzed the ESEB 2022 (Brazilian Electoral Study) database, a post-electoral survey conducted since 2002. ESEB is a project associated with the Comparative Study of Electoral Systems Project (CSES) at the University of Michigan, with participation from dozens of institutions from various countries.

April 2023 | Social Network Analysis - Nova SB

Analyzed and produced a report on the federal government's image on social media, including the construction of network graphs in Gephi for mapping connections and interaction dynamics.

October 2022 | Territories in Network - Aprendiz and Fundação Vale

Developed automation and project management products, constructed a database, and produced analyses on school dropout in 16 municipalities across the country.

September 2022 | VotoMaps - MAF dataScience

Developed online electoral reports and maps for candidates running for elective office (<https://www.votomaps.com/>). Programming was done in JavaScript using the Mapbox library.

March 2022 | School Dropout and Child Labor - Aprendiz

Conducted quantitative data collection and analysis on school dropout and child labor in the city of São Paulo during the pandemic.

April 2022 | Campaign Consulting - Federal Deputy Candidate Gabriela Manssur

Performed quantitative and geocoded data collection and analysis to support the electoral campaign for a federal deputy candidate in São Paulo.

December 2021 | Police Candidates - University of Bradford

Conducted electoral and territorial analysis of candidacies by civil police, military police, and vigilantes in the State of São Paulo during the municipal elections from 2004 to 2020.

November 2021 | Qualitative and Quantitative Research - Grupo Mais

Conducted qualitative focus group research and quantitative online research with Banco do Brasil employees. The research aimed to gather insights to guide Grupo Mais's electoral campaign in 2022 for the PREVI (BB's Pension Fund) board.

September 2021 | Electoral Mapping - State Deputy Candidate

Performed geographical analysis of candidate votes in 2018 and 2020 in the state of São Paulo.

August 2021 | Telephone Survey with APEOESP Associates - DIEESE

Conducted telephone surveys, data analysis, and programmed an Android auto-dialing application.

August 2021 | Digital Marketing - Political Party

Developed an algorithm for WhatsApp and email communication to invite members to a party event.

August 2021 | Online Research Database - Private Project

Analyzed and produced a report for an online research project on individual economic behavior. This research was produced for a master's thesis.

March 2021 | Belém do Pará 2020 Elections - Perspectiva Comunicação

Collected and analyzed electoral data in Pará from 2016 to 2020. Performed geospatial data analysis of mayoral candidate votes in Belém in 2020. Analyzed electoral poll results in Belém and Santarém in 2020.

November 2020 | 2020 Electoral Data - Concordia Public Affairs

Developed a database on mayoral and councilor votes in 2020 across the country. Produced a report on the number of elected officials, mayoral re-election rates, election by gender, and analysis of fragmentation in municipal legislatures.

April 2020 | COVID-19 Observatory - Miami University

Assembled and analyzed data on state government actions in combating COVID-19. Established the COVID-19 Observatory for Latin America.

September 2019 | Twitter Web Scraping - POLBRAS/UNICAMP

Developed a database on party mentions on Twitter.

August 2019 | Telephone Survey with SINDMETALSJC Associates - DIEESE

Conducted telephone surveys with associates of the Metalworkers' Union of São José dos Campos. Performed data analysis and programmed an auto-dialing application in Kotlin language.

June 2019 | Forced Labor Database - FGV

Assembled and analyzed a database on forced labor in the country.

March 2019 | Electoral Mapping - City Councilor Candidate

Performed geographical analysis of a PT city councilor candidate's votes in 2016 in Campinas.

September 2018 | FAPESP Web Scraping - ADUNICAMP

Developed a Web Scraping algorithm to build a database on FAPESP scholarship recipients. Conducted data analysis and composed a report.

Education

- **Postdoctoral Research** | EAESP/FGV | São Paulo, SP Since January 2024 Researcher at the Center for Public Sector Politics and Economics (CEPESP).
- **Sandwich Doctorate** | Massachusetts Institute of Technology (MIT) | Cambridge, MA, USA February to June 2020 Focus on advanced quantitative methods and Machine Learning.
- **Ph.D. in Political Science** | UNICAMP | Campinas, SP January 2016 – August 2022 Dissertation on partisan fragmentation in Brazil.
- **Master's in Political Science** | UNICAMP | Campinas, SP January 2014 – December 2015 Thesis on the electoral geography of "bullet caucus" candidates.
- **Bachelor's in Social Sciences** | USP | São Paulo, SP January 2003 – December 2008